M.KUMARASAMY COLLEGE OF ENGINEERING

EEB1223 – DESIGN THINKING

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TITLE : Automated Fault Recovery Fuse Carrier for Rural Distribution Network

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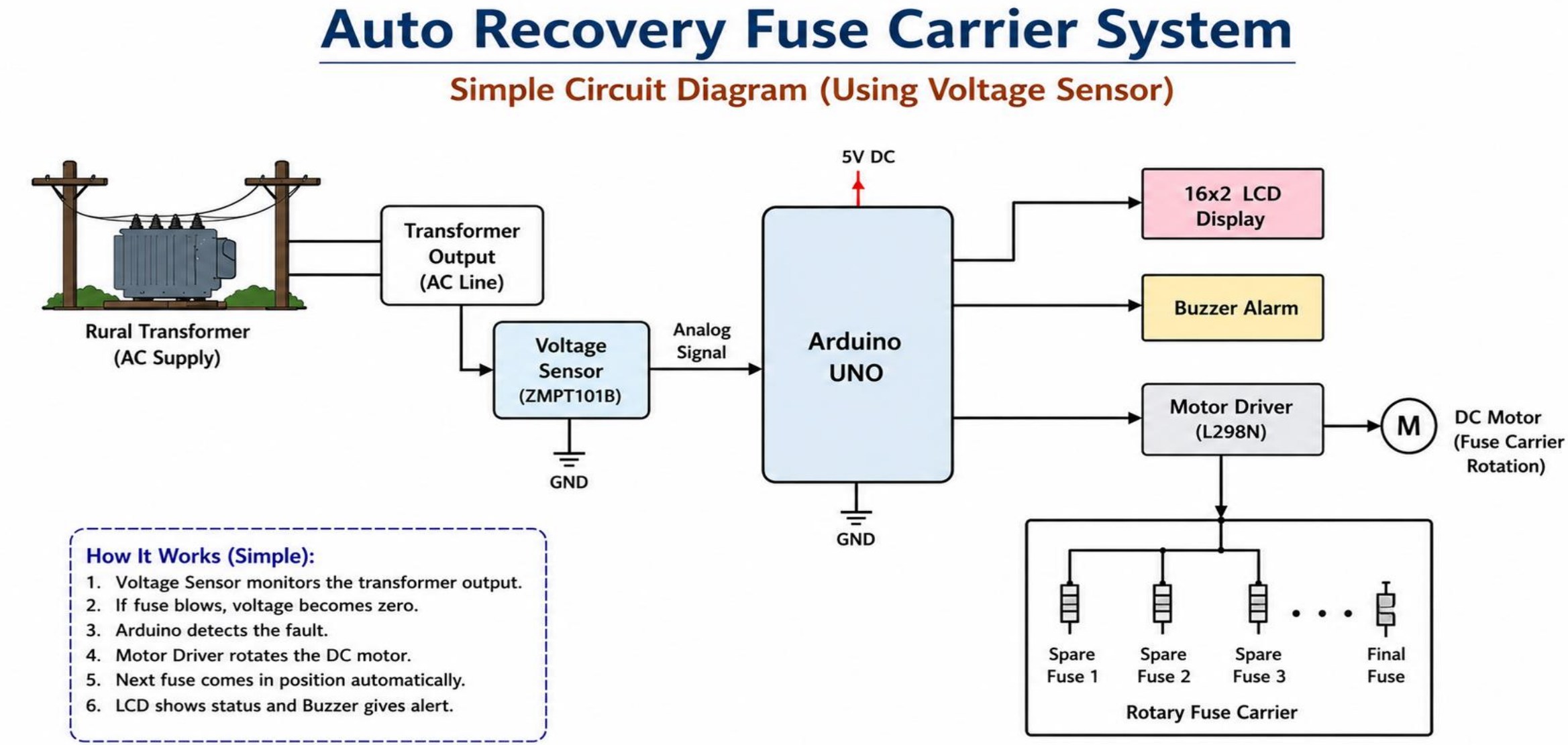
ABSTRACT

This project presents an Auto Recovery Fuse Carrier System designed for rural transformers to reduce power interruptions caused by blown fuses. A current sensor continuously monitors the load, and when a fault is detected, the signal is sent to an Arduino microcontroller. The controller then activates a motor driver, which runs a DC motor to rotate a fuse carrier containing multiple spare fuses. This automatically replaces the damaged fuse without human intervention. The system helps in reducing downtime and ensures a more reliable power supply. It is especially useful in remote areas where maintenance is difficult and time-consuming.

Proposed objective

The main objective of this project is to design and develop an Auto Recovery Fuse Carrier System for rural transformers to ensure continuous power supply. The system aims to automatically detect faults using a current sensor and respond quickly through an Arduino microcontroller. It focuses on replacing blown fuses by rotating a fuse carrier using a motor-driven mechanism without human intervention. This helps to reduce power outages, improve system reliability, and minimize maintenance efforts. The project also aims to provide a cost-effective and efficient solution suitable for remote and rural areas. Additionally, it enhances safety by reducing the need for manual handling of high-voltage components. It also supports faster fault recovery, ensuring uninterrupted service to consumers.

Proposed Architecture



COMPONENTS

- **Power Supply Operation:**

  The power supply provides 5V/12V to all components in the system for proper functioning. It ensures stable and continuous power for sensing, control, and motor operations.

- **Voltage Sensing Operation:**

  The voltage sensor monitors the transformer voltage continuously. It detects voltage loss when the fuse blows.

- **Signal Conditioning Operation:**

  The signal conditioning circuit processes the raw signal from the sensor. It filters noise and converts it into a suitable form for the Arduino.

- **Fault Detection & Control Operation (Arduino):**

  The Arduino microcontroller analyzes the conditioned signal to detect faults. It then sends control signals to the motor driver and output devices.

COMPONENTS

- **Motor Driving Operation:**

  The motor driver receives signals from the Arduino and controls the DC motor. It regulates the motor's direction and movement to perform fuse replacement.

- **Fuse Replacement Operation:**

  The DC motor rotates the rotary fuse carrier when a fuse blows. This automatically connects the next available fuse to restore power.

- **LCD Display Operation:**

  The LCD displays the fuse condition such as "Fuse Replaced" or "Final Fuse Replaced." It helps users monitor system status in real time.

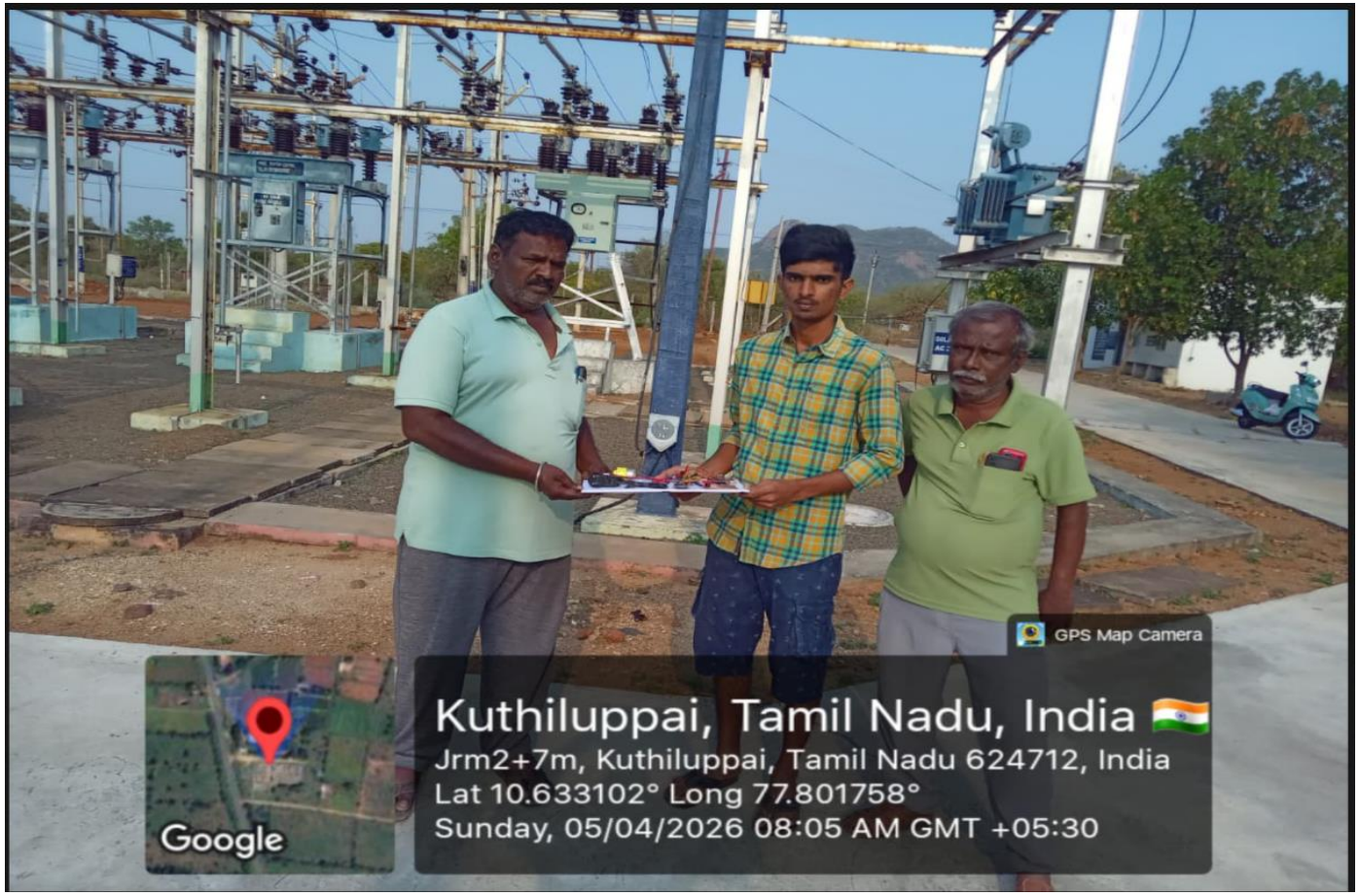
- **Buzzer:**

  The buzzer is used to provide an alert when the fuse fails and no voltage is detected in the fuse carrier. It gives a continuous sound to indicate the fault condition and helps in identifying the fuse replacement process.

EXISTING AND PROPOSED

EXISTING	PROPOSED
The existing electrical safety system mainly uses manual fuse protection and basic circuit breakers. It does not provide early leakage detection or automatic response. During rainy seasons, faults may go unnoticed, increasing the risk of electric shock and fire accidents.	The proposed system uses Arduino with ACS712 sensor to detect current leakage automatically. It disconnects the power using a relay and activates buzzer and LEDs for warning. It also displays status on LCD and can automatically move the fuse using a motor mechanism for improved safety.

REAL TIME IMAGE



CONCLUSION

The Automated Fault Recovery Fuse Carrier for Rural Distribution Network provides an efficient and reliable solution to improve power supply continuity in rural areas. The system automatically detects fault conditions such as overcurrent or short circuit and isolates the faulty line using a controlled mechanism. By integrating sensors, a microcontroller, relay module, and motorized fuse carrier, the system reduces the need for manual inspection and replacement.This automation minimizes downtime, improves safety for line workers, and ensures faster power restoration to consumers. The use of low-cost components makes it suitable for rural distribution networks where maintenance resources are limited. Overall, the proposed system enhances reliability, safety, and operational efficiency in rural power distribution systems.

THANK YOU